Namma Space: Autonomous 3D Indoor Radiance Fields from Smartphone Video for Zero-Install Spatial Web Roaming

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1. Problem Statement & Project Objective

Navigating real estate interiors remotely has historically suffered from a stark dichotomy: cheap but disjointed 2D photograph carousels and pre-rendered 360-degree panoramas versus photorealistic but operationally friction-heavy LiDAR scanning rigs. Panorama solutions confine users to fixed discrete camera bubbles ('bubble hopping'), inducing severe motion parallax disorientation and precluding continuous natural exploration. Conversely, specialized LiDAR rigs (e.g. Matterport Pro3) require multi-thousand-dollar hardware and tedious multi-station setups. **The Objective of Namma Space:** To construct an autonomous, end-to-end photogrammetric twin pipeline that converts standard consumer smartphone walkthrough videos into photorealistic, continuous 3D radiance fields. The reconstructed space must be instantly navigable in any modern web browser via first-person WASD fly controls and interactive mouse look at steady 60 FPS, without requiring user authentication, native plugins, or client-side software installations.

2. System Architecture & Dual-Tier Topology

Namma Space strictly decouples high-performance geometric reconstruction from ultra-lightweight client-side spatial inference across four architectural tiers:

Architectural Layer	Engine / Component	Key Functional Responsibilities & Boundary
1. Data Ingestion	Smartphone Camera (iOS/Android)	1080p @ 30 fps, 1.0x wide lens, AE/AF locked, 4-stage dual-loop walking route.
2. Reconstruction Worker	Pipeline Worker + PyColmap	Single-camera calibration, Covariant SIFT, 0% discardation Sim3d alignment, SOR floater purge.
3. Web Infrastructure	Vanilla HTML5 / Modern CSS	Dark-mode UI, in-browser upload portal (/api/upload), live stepper, zero-backend static deployment.
4. First-Person Engine	GaussianSplats3D + Three.js	WebGL2 radiance field rendering, WebAssembly multi-threaded radix depth sort, WASD telemetry HU

3. High-Level Data Transformation Flow

- **Direct Web Ingestion:** Users drag-and-drop raw smartphone video walkthroughs directly into the web interface (`/web/index.html`), streaming chunks to an asynchronous processing worker with real-time SSE progress telemetry.
- **Autonomous 3D Spatial Solving:** The pipeline runs headless, extracting keyframes, solving camera poses across sharp turns, purging floaters, and synthesizing dense oriented surfels into binary `.splat` format.
- **Zero-Latency Web Delivery:** The generated 3D scene streams immediately to the client canvas with auto-framing, enabling instant walkthrough exploration at 60 FPS without page reload.

4. Reconstruction Pipeline Design & Algorithmic Foundations

Transforming unconstrained handheld video into continuous radiance fields requires solving significant mathematical challenges: motion blur during rapid turns, focal length drift, and floating triangulation artifacts. Namma Space deploys a multi-stage photogrammetric pipeline:

- **Temporal Subsampling & Frame Ingestion:** Handheld video at 30 fps possesses severe redundancy and inter-frame blur. OpenCV temporally samples frames at 2.5 fps (~100–150 frames for a 60s walk) while computing Laplacian variance blur scores to reject smeared transition frames. Frames are normalized to 848x478 for optimal speed-to-accuracy balance.
- **Unified Physical Sensor Calibration (`CameraMode.SINGLE`):** Assigning independent camera IDs forces SfM solvers to estimate focal lengths per frame, triggering divergence during rapid camera turns. Namma Space enforces a single shared physical camera matrix (`SIMPLE_RADIAL`), locking the focal length ($f = 1017.6\,\text{px}$) and principal point to eliminate focal drift.
- **Covariant SIFT Feature Extraction:** To retain tracking across motion blur and dark room sections, the extractor operates with 2x octave upsampling (`first_octave = -1`), high-sensitivity peak threshold (0.004), edge threshold (15.0), and domain-size pooling (DSP). This detects 800–4,000 robust features per frame, bridging previously severed turning sequences.
- **Chronological Pair Matching & Loop-Closure Graph:** Rather than naive sequential pairing susceptible to database ID reordering, the system constructs an explicit graph matching each frame t with local neighbors ($t \pm 1..25$) plus quadratic long-range jumps ($t \pm 35, 50, 75, 100, 150, 200, 250, 300, 350$) for robust loop closure (12,400+ verified pairs).
- **0% Discardation Multi-Model SfM & Closed-Form Sim3d Alignment:** When fast turns split reconstructions into submodels, traditional tools discard non-primary segments, dropping up to 70% of the room. Namma Space retains all submodels and fuses them using exact closed-form similarity transformations (Sim3d: 7-DoF scale s, rotation $\mathbf{R}$, translation $\mathbf{t}$) computed via `align_reconstructions_via_proj_centers`, achieving zero dead-reckoning drift.
- **Statistical Outlier Removal (SOR) Floater Purge:** Multi-view consensus rejects all points with track length < 3. A k-d tree Statistical Outlier filter ($k=12$) computes local mean neighbor distances μ and standard deviation σ, pruning points exceeding $\mu + 1.25\sigma$ (purging 43,000+ noisy air points).
- **Tight Voronoi Pancake Surfel Synthesis:** Spherical Gaussians cause blurry smog when large ($0.50\,\text{m}$) and black out when zooming. We scale radii strictly to $s_{xy} = \text{clip}(\text{avg_dist} \times 0.40, 0.003, 0.045)\,\text{m}$ (3mm to 4.5cm) and flatten along surface normals ($s_z = 0.15 \cdot s_{xy}$) with high opacity ($\alpha \ge 235$), yielding razor-sharp solid geometry.

Pipeline Stage	Algorithmic Method	Mathematical Operation / Key Parameters	Output Artifact
Frame Ingestion	Temporal Subsampling	2.5 fps sampling, Laplacian blur filter	JPEG Keyframe Sequence
Camera Intrinsics	Shared Physical Sensor	CameraMode.SINGLE, SIMPLE_RADIAL, f=1017.6px	Unified Camera Model
Feature Detection	Covariant SIFT + DSP	first_octave=-1, peak_thresh=0.004, edge=15.0	128D SIFT Descriptors
Feature Matching	Chronological Pair Graph	Local window [t-25..t+25] + quadratic loop jumps	two_view_geometries (DB)
Structure-from-Motion	Incremental Mapping	abs_pose_min_inliers=12, init_tri_angle=3.0 deg	Multi-Model Sparse Point Cloud
Submodel Fusion	Sim3d Projection Centers	Closed-form Sim3d (s, R, t) alignment across overlaps	Unified Global Coordinate Space
Outlier Rejection	Multi-View Consensus + SOR	track_length >= 3, k=12, dist <= mu + 1.25*sigma	Cleaned Surfel Cloud
Surfel Synthesis	Pancake Splat Parameterization	s_xy in [0.003, 0.045]m, s_z = 0.15*s_xy, alpha >= 235	scene.splat (32-byte binary)

5. Smartphone Capture Standard Operating Procedure (SOP) Details

Photogrammetric triangulation relies strictly on multi-view baseline parallax. Pure rotational panning (standing in one spot and turning) produces zero baseline displacement, resulting in catastrophic epipolar degeneracy. Our standardized 4-stage mobile capture protocol guarantees reliable 3D reconstruction:

- **Stage 1 — Environmental Conditioning:** Turn on all artificial room lighting to maximize photon count and minimize sensor gain noise. Close window curtains/blinds to eliminate extreme specular sunlight blooms. Open all interior doors. For featureless walls, temporarily position textured visual cues (e.g., sticky notes or posters) to provide reliable keypoint tracking.
- **Stage 2 — Sensor Configuration & Freezing:** Hold smartphone strictly in landscape orientation. Select standard 1.0x optical wide lens (strictly avoid 0.5x ultra-wide due to unmodeled non-linear radial distortion). Set resolution to 1080p @ 30 fps. **Mandatory AE/AF Lock:** Long-press on the center of the room to permanently lock auto-exposure, auto-white-balance, and focal distance across the entire video.
- **Stage 3 — Dual-Loop Walking Trajectory:** Execute a deliberate 45–75 second heel-to-toe 'ninja walk' (0.5–0.8 m/s):
  - *Outer Loop (Perimeter Pass):* Walk along the room perimeter walls facing inwards at a 45-degree angle, sweeping baseboards, corners, and wall-floor junctions (~30–40s).
  - *Inner Loop (Central Furniture Orbit):* Transition smoothly into a tighter circular orbit around central furniture (desks, sofas, beds) to capture occlusion geometry (~20s).
- **Stage 4 — Stationary Buffer Holds:** Maintain a completely motionless 3–4 second hold at the exact beginning and end of recording to isolate touchscreen tapping vibrations from keyframe selection.

Capture Checkpoint	Operational Protocol	Failure Mode if Ignored	Corrective Action
Exposure & Focus	Mandatory AE/AF Lock prior to recording	Photometric flickering, SIFT descriptor mismatch	Long-press smartphone screen to lock exposure.
Lens Selection	Standard 1.0x optical wide lens	0.5x ultra-wide introduces severe barrel distortion	Use 1.0x default camera profile.
Walking Gait	Heel-to-toe 'ninja walk' (0.5-0.8 m/s)	Vertical rolling-shutter wobble, motion blur	Glide steadily without abrupt directional jerks.
Trajectory Route	Dual-loop: perimeter pass + central orbit	Occluded dead zones behind furniture	Ensure central objects receive 360-degree coverage.
Monochromatic Walls	Add high-contrast temporary markers	Zero keypoint detection, lost camera tracks	Place textured sticky notes on plain surfaces.

6. Technical Approach & Architectural Trade-off Analysis

Design Dimension	Adopted Solution	Evaluated Alternatives	Architectural Rationale & Justification
Spatial Representation	3D Gaussian Radiance Field	NeRF, Textured Triangle Mesh	3DGS renders in real-time via tile rasterization; NeRF requires slow ray-marching (1-
Multi-Model Fusion	Closed-Form Sim3d Alignment	Velocity Dead Reckoning, Discardation	Sim3d provides exact metric scale & rotation alignment; dead reckoning drifts expon
Web Delivery Engine	WebGL2 + WebAssembly Worker	Server-Side Video Streaming (Pixel Stream)	Client-side WebGL2 eliminates expensive cloud GPU streaming costs ($0.00 server
UI Architecture	Vanilla HTML5 / CSS / ES Modules	React / Next.js / Angular SPA	Eliminates hydration lag, bundle bloat, and build steps; loads instantly on consumer

7. Browser Viewer Engine & First-Person WASD Navigation

The Namma Space client delivers an uncompromised spatial computing experience directly in standard web browsers:

- **First-Person WASD Roaming:** Keyboard event listeners compute real-time translational displacement along the camera's view-look and strafe vectors: $\Delta \mathbf{P}_{\text{fwd}} = \mathbf{V}_{\text{look}} \cdot v \cdot \Delta t$ and $\Delta \mathbf{P}_{\text{strafe}} = (\mathbf{V}_{\text{look}} \times \mathbf{V}_{\text{up}}) \cdot v \cdot \Delta t$. The camera up-vector is fixed to $[0, -1, 0]$ to prevent disorientation.
- **Interactive Mouse Orbit:** Left-click drag controls spherical pitch/yaw; right-click drag pans laterally; scroll wheel modulates zoom.
- **Live Telemetry HUD:** A glassmorphic HUD renders real-time coordinate telemetry (x, y, z) and active keypress indicators.
- **Asynchronous Web Worker Radix Depth Sorting:** To prevent z-fighting across semi-transparent Gaussians, splats are dynamically sorted by camera depth inside a dedicated Web Worker via WebAssembly, sustaining steady 60 FPS on integrated APUs.
- **Client-Side Cross-Origin Isolation Bridge:** Static hosts (GitHub Pages) do not permit custom headers. Namma Space deploys `coi-serviceworker.js` to dynamically register COOP (`same-origin`) and COEP (`credentialless`) on the client side, unlocking `SharedArrayBuffer`.

8. Empirical Results & Performance Benchmarks

Verification was conducted using automated Chrome DevTools Protocol (CDP) test suites across both real-world smartphone video and synthetic benchmarks:

Benchmark Metric	Measured Real-World Value	Verification Evidence & Audit Status
3D Radiance Model Size	12.12 MB (scene.splat)	Binary 32-byte format; complete spatial twin (PASS).
Registered Camera Count	58 Walkthrough Views	Continuous indoor trajectory with initial overview hero pose (PASS).
Total Gaussian Elements	378,886 3D Gaussians	Dense photometric coverage preserving white walls, bed, & floor (PASS).
Initial Stream & Parse	180ms localhost (1.9s 4G)	Instant buffer construction without hydration overhead (PASS).
Rendering Frame Rate	Steady 60.0 FPS	Verified WebGL2 frame telemetry under continuous WASD flight (PASS).
Hero Overview Spawn	x: 0.470, y: -0.467, z: -5.779	Spawns facing full interior overview with guitar and bedding (PASS).
Console Error Audit	0 Exceptions, 0 Warnings	CDP Runtime.consoleAPICalled verified 100% clean stream (PASS).

9. Reproducibility, Attribution & Statement of Originality

- **Local Reproduction:** Clone repository, install dependencies in Python virtual environment, and run `python serve_web.py`. Access `http://127.0.0.1:8080/index.html` to test live upload and viewer roaming.
- **Third-Party Attributions:** Inria / MPI (3D Gaussian Splatting, Kerbl et al., SIGGRAPH 2023); COLMAP (Schönberger & Frahm, CVPR 2016); GaussianSplats3D (Mark Kellogg et al., WebGL2 engine); Three.js (Cabello et al., 3D math & scene graph).

Statement of Original Implementation: Namma Space independently developed: (1) The responsive dark-mode web portal with in-browser video upload and real-time SSE progress telemetry; (2) The 0% discardation multi-model Sim3d closed-form similarity alignment engine; (3) The k-d tree Statistical Outlier Removal (SOR) floater purging filter; (4) The tight Voronoi pancake surfel parameterization ($0.003–0.045\,\text{m}$); (5) The client-side Cross-Origin Isolation service worker bridge; (6) The 6-DoF first-person WASD roaming engine with live coordinate telemetry HUD; and (7) The 47-gate automated compliance audit test suite.